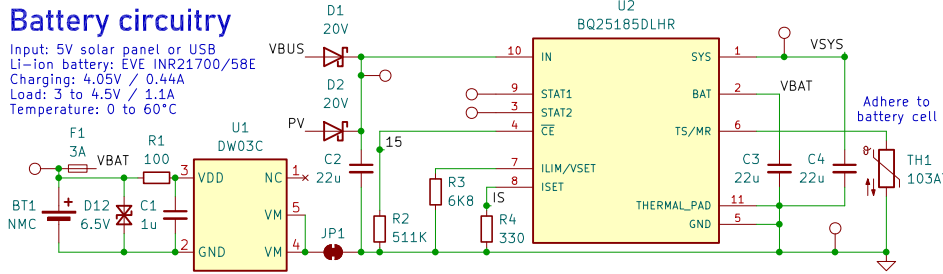


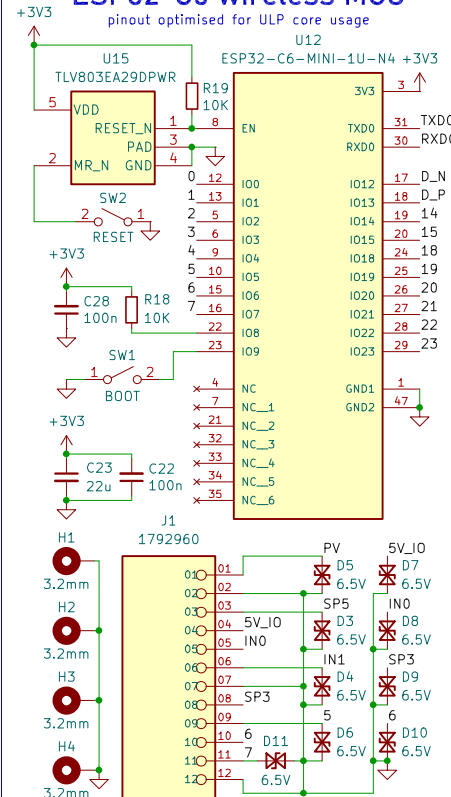
Battery circuitry

Input: 5V solar panel or USB
Li-Ion battery: EVE INR21700/58E
Charging: 4.05V / 0.44A
Load: 3 to 4.5V / 1.1A
Temperature: 0 to 60°C

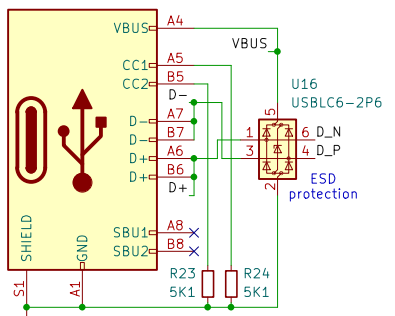


ESP32-C6 wireless MCU

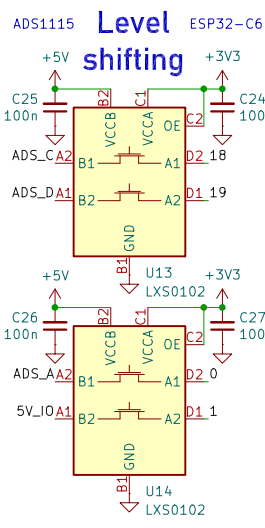
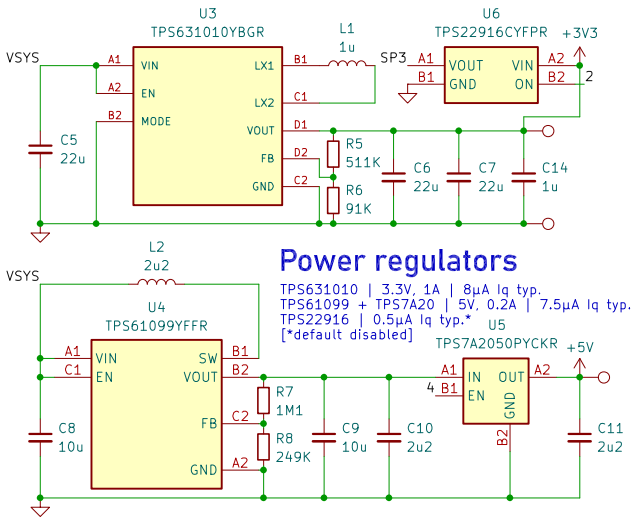
pinout optimised for ULP core usage



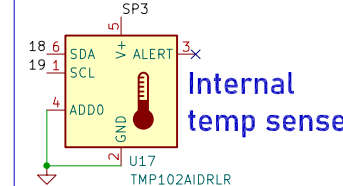
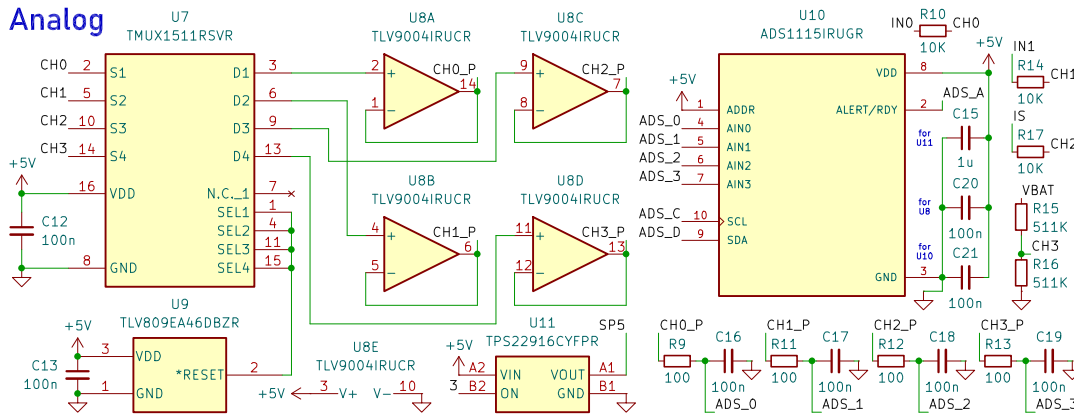
USB Type-C



Power regulators



Analog



OpenTank

Most ESP32 development boards are simply unsuitable for outdoor sensor monitoring, with idle current consumption in the mA range. Efficient solar battery charging is seldom integrated on-board, and other forms to optimise runtime are not considered enough. OpenTank aims to solve this. It includes a 5V ADC with added powered-off protection.

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Title: Low-power ESPHome outdoor node for 5V analog sensor

Size: User Date: 2026-02-17
KiCad E.D.A. 8.0.1

Rev: B
Id: 1/1